Presentation 2

What did each of the team members do?

Henry: Mandatory programming tasks 1-4, 8 and task 11 (patch imagefilename standardization, inspect data quality, inspect distribution of slices, create new cases from cases with too many slices in a stain)

Vicky: two moderate tasks 5 and 6, add entropy regularization for attention scores to avoid peaky attention scores, and use gated attention instead of the current one.

Zijin: Visualization and model evaluation (task 12&13). I computed effective attention scores, and generated visualizations including training/validation loss curves, deviation plots, and attention distributions. I compared correctly and incorrectly classified cases, identified attention concentration patterns, and extracted representative cases for qualitative inspection. Lastly, I proposed possible next steps such as data argumentation.

How does it help the project?

Henry: It creates a more robust data input that identifies unusual files and data exploration in python notebook finds the distribution of every slice in a stain.

Vicky: The newer model increases test accuracy to 0.7895, decreases test loss to 0.5891.

Gated Attention improves accuracy by introducing nonlinearity, allowing the model to capture complex diagnostic patterns that simple linear attention cannot detect.

Entropy Regularization improves stability and lowers loss by preventing the model from concentrating too heavily on a few noisy patches, encouraging a more balanced evaluation of the entire tissue slide.

Zijin: formalized a reusable attention-distribution framework that allows us to quantitatively evaluate how the model allocates evidence when improving models. Also, I extracted some key cases (both correct and incorrect predictions) that show distinct attention patterns. These cases can serve as focused study examples for future model refinement and potential training adjustments.

Issues faced that haven't been resolved:

- Current pseudocase function works but the code is messy and could be improved to avoid data leakage
- Training currently runs on CPUs, taking ~15 minutes per epoch. This limits extensive hyperparameter tuning (e.g., broader λ search) and efficient 5-fold cross-validation.
- The current attention analysis clearly shows where the model is focusing, but does not fully explain why the model makes errors. Eg. Requiring

additional spatial context or cross-stain comparative analysis beyond the current visualization outputs.

Proposed next steps:

- New results (was not ready in time for recording) look very good but could be deceptive. Need to investigate the splits. Also create a better solution to view what the number of slices are for each case after data augmentation and stratified splitting.
- Use GPU instead of CPU to train models.
- Task 14: visualize patch attention on tissue slice.

References

Henry: KimiaNet Github↗

Vicky: Gated Attention+Entropy Github

Zijin: presentation 4 of Fall Quarter, GenAI Tools, Fall 2025 MIL Trainer Final Model and Analysis

Links to the presentation & code:

- **Presentation:** **Presentation2_Team6**
- **Code:** **Code2_KimiaNet_10Feb**